

Exercise 1: Mocking a Service Dependency in a Controller Test

```
import static org.mockito.Mockito.when;

import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;
import org.junit.jupiter.api.Test;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.autoconfigure.web.servlet.WebMvcTest;
import org.springframework.boot.test.mock.mockito.MockBean;
import org.springframework.test.web.servlet.MockMvc;

@WebMvcTest(UserController.class)
public class UserControllerTest {

    @Autowired
    private MockMvc mockMvc;

    @MockBean
    private UserService userService;

    @Test
    public void testGetUser() throws Exception {

        User user = new User();
        user.setId(1L);
        user.setName("John");
        when(userService.getUserById(1L)).thenReturn(user);
        mockMvc.perform(get("/users/1"))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.id").value(1))
            .andExpect(jsonPath("$.name").value("John"));
    }
}
```

Output:

T E S T S

Running UserControllerTest

Tests run: 1

Failures: 0

Errors: 0

BUILD SUCCESS